
Supplemental Material for “Beyond Quantum Mechanics: Local QM SpaceTime Algebra Prediction of the Singlet State”

Validation of the Local QM Product Calculation Prediction replacing Pauli Matrices with SpaceTime Algebra and 2D Vectors, Based on Joy Christian’s 3-Sphere Model.

This makes the calculation real; no probabilities. <https://einstein-physics.org/>

Local Quantum Mechanical Prediction of the Singlet State Using SpaceTime Algebra

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Caveat: This is a simulation of a QM prediction; it is not a simulation of an experiment.

Load Clifford Package, Set Run Time Parameters, Initialize Arrays

```
In[ ]:= << "clifford.m"
Clear[$SetSignature];
$SetSignature = {1, 3}; (*Signature set for SpaceTime Algebra*)
γ0 = e[1]; γ1 = e[2]; γ2 = e[3]; γ3 = e[4];
σ1 = GeometricProduct[γ1, γ0]; (*SpaceTime Bivectors*)
σ2 = GeometricProduct[γ2, γ0];
σ3 = GeometricProduct[γ3, γ0];
I4 = Pseudoscalar[4];
m = 100000; (*Number of trials*)
bins = 721; (*Number of degrees for averaging*)
s1 = s2 = qpc = ConstantArray[0, m];
aa = bb = gA = gB = A = B = th = pc = ConstantArray[0, m];
sumAB = countAB = ConstantArray[0, bins];
phi = (1 / Sqrt[2]) (1 - GeometricProduct[I4, σ2]); (*STA Particle Wavefunction *)
chi = (1 / Sqrt[2]) (GeometricProduct[I4, σ2] - 1);
(*STA Particle Wavefunction *)
```

Generating Particle Data with Three Independent Do-Loops

```
In[ ]:= Do[s1[[h]] = RandomPoint[Sphere[]].{σ1, σ2, σ3};
(*Uniform Unit 3D Bivectors*) (*Spin bivector to A*)
s2[[h]] = -s1[[h]], {h, m} (*s1+s2 = 0,
conservation of angular momentum*) (*Spin bivector to B*)

In[ ]:= Do[α = RandomReal[{0, 2 π}];
a = {Cos[α], Sin[α], 0}.{σ1, σ2, σ3};
aa[[h]] = α;
stA = Expand[GeometricProduct[Turn[phi], a, s1[[h]], phi]];
(*Particle-Detector Interaction*)
cosas1 = Grade[stA, 0];
gA[[h]] = stA;
A[[h]] = Sign[cosas1], {h, m}]
```

```

In[ ]:= Do[ $\beta$  = RandomReal[{0, 2  $\pi$ }] ;
  b = {Cos[ $\beta$ ], Sin[ $\beta$ ], 0}.{ $\sigma$ 1,  $\sigma$ 2,  $\sigma$ 3};
  bb[[h]] =  $\beta$ ;
  stB = Expand[GeometricProduct[Turn[chi], s2[[h]], b, chi]];
  (*Particle-Detector Interaction*)
  cosbs2 = Grade[stB, 0];
  gB[[h]] = stB;
  B[[h]] = Sign[cosbs2], {h, m}]

```

Verification of the Local QM Product Calculation Prediction

```

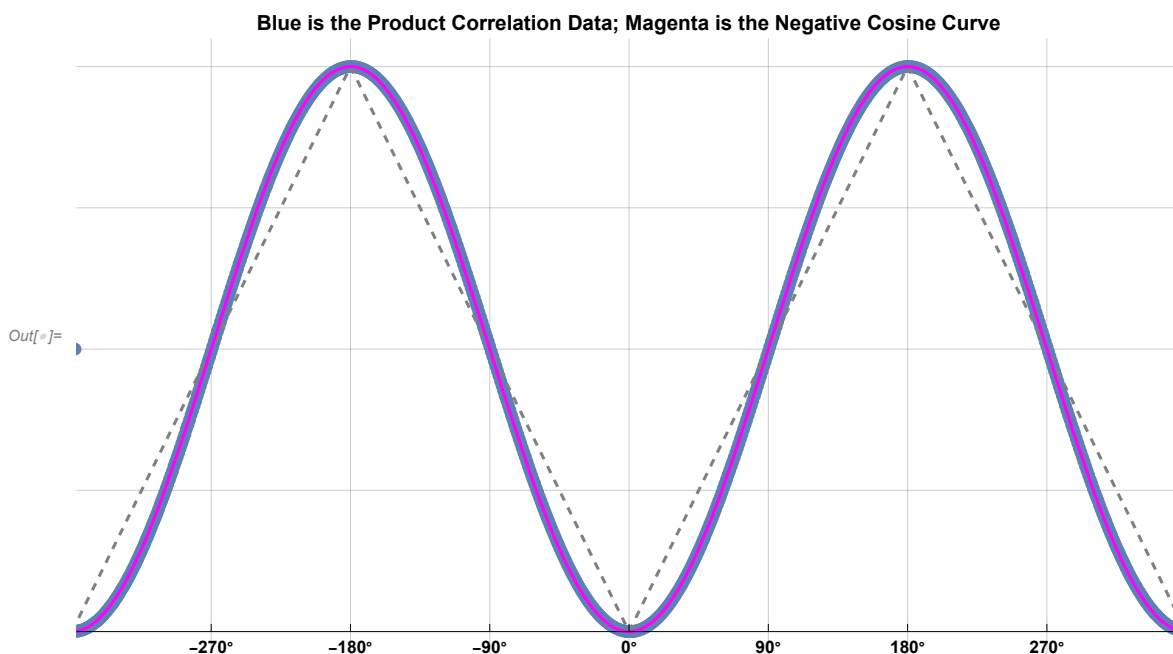
In[ ]:= Do[(*STA Product Calculation*)
  gpc = Grade[Expand[GeometricProduct[gA[[h]], gB[[h]]], 0];
  pc[[h]] = Grade[Expand[GeometricProduct[gA[[h]], gB[[h]]], 2];
  (*Cross Product Extraction*)
  a1 = aa[[h]];
  b1 = bb[[h]];
  angle = (b1 - a1) / Degree + 361;
  theta = Ceiling[angle];
  sumAB[[theta]] += gpc;
  countAB[[theta]] += 1, {h, m}]
Eavg = RotateLeft[N[sumAB / (countAB + 10^-7)], 1];
(*Average of Product Calculation Scalar Per One Degree Increments*)

```

```

In[ ]:= simulation = ListPlot[Eavg, PlotRange → {{3, 717}, {-1.1, 1.1}},
  PlotMarkers → {Automatic, Small}, AspectRatio → 9 / 16,
  Ticks → {{0, "-360°"}, {90, "-270°"}, {180, "-180°"}, {270, "-90°"}, {360, "0°"},
    {450, "90°"}, {540, "180°"}, {630, "270°"}, {720, "360°"}}, Automatic},
  GridLines → Automatic, AxesOrigin → {0, -1.0}, ImageSize → Large, PlotLabel →
    "Blue is the Product Correlation Data; Magenta is the Negative Cosine Curve",
  LabelStyle → Directive[Bold, Black]];
negcos = Plot[-Cos[x Degree], {x, 1, 720}, PlotStyle → {Magenta}];
p1 = Plot[-1 + 2 x1 Degree /  $\pi$ , {x1, 0, 180}, PlotStyle → {Gray, Dashed}];
p2 = Plot[3 - 2 x2 Degree /  $\pi$ , {x2, 180, 360}, PlotStyle → {Gray, Dashed}];
p3 = Plot[-1 + 2 (x3 - 360) Degree /  $\pi$ , {x3, 360, 540}, PlotStyle → {Gray, Dashed}];
p4 = Plot[3 - 2 (x4 - 360) Degree /  $\pi$ , {x4, 540, 720}, PlotStyle → {Gray, Dashed}];
Show[simulation, p1, p2, p3, p4, negcos]

```



Computing Averages

```

In[ ]:= AveA = N[Total[A] / m];
AveB = N[Total[B] / m];
Print[" <A> = ", AveA, " <B> = ", AveB];
meanpc = Expand[Mean[pc]];
Print["Cross products vanish, meanpc = ", meanpc];

<A> = -0.0012 <B> = 0.00756
Cross products vanish, meanpc =  $-8.61614 \times 10^{-20} e_2 e_3 - 2.22847 \times 10^{-19} e_2 e_4 - 0.000490095 e_3 e_4$ 

```

The A and B averages of the +/- 1 outcomes are to demonstrate that they match the quantum mechanical prediction.